RACE
Regular Analysis Through Continuous Exercise

The s-Block Elements : RACE-01

Group 1 Elements; Alkali Metals, Electronic Configuration, Atomic and Ionic Radii, Ionisation Enthalpy, Hydration Enthalpy, Physical Properties

1. Which of option is correct for density of alkali metals.
(A) Na > K (B) K > Na (C) Na = K (D) None of these

CSB082

2. Which option is incorrect.
(A) Cs gives blue colour to oxidising flame
(B) Li gives crimson-red colour to the oxidising flame
(C) K gives brick–red colour to the oxidising flame
(D) Na gives golden yellow colour to the oxidising flame.

CSB083

3. Which option is correct for 1st Ionization potential.
(A) Li > K (B) Na > K (C) Rb > Cs (D) All of these

CSB084

4. Atomic radii of alkali metals (M) follow the order : Li < Na < K < Rb but ionic radii in aqueous solution follow the reverse order $Li^{\oplus} > Na^{\oplus} > K^{\oplus} > Rb^{\oplus}$. The reason for the reverse order is –
(A) Increase in the ionisation energy
(B) Decrease in the metallic bond character
(C) Increase in the electropositive character
(D) Decrease in the amount of hydration

CSB087

Bankers Academy JEE

Chemical properties of Alkali metals (Reactivity towards Air, Water, Halogens, Dihydrogen, Reducing Nature, Solutions in Liquid NH_3)

5. When K_2O is added to water, the solution becomes basic in nature because it contains a significant concentration of
(A) $K^{\oplus}$ (B) $O^{2\ominus}$ (C) $OH^{\ominus}$ (D) $O_2^{2\ominus}$

CSB079

6. Which option is correct for alkali metals.
(A) Alkali metals are paramagnetic and their compounds are diamagnetic.
(B) Alkali metals are paramagnetic and their compounds are also paramagnetic.
(C) Alkali metals are diamagnetic and their compounds are paramagnetic.
(D) Alkali metals are diamagnetic and their compounds are diamagnetic.

CSB086

Uses of Alkali Metals

7. Which option is correct?
(A) Lithium is store in wax (B) Na, K, are stored in kerosene.
(C) Rb, Cs are stored in kerosene (D) All of these.

CSB085

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General Characteristic of Compounds of Alkali Metals (Oxides and Hydroxides, Halides, Salts of Oxo-acids)

8. Potassium superoxide is used in.
(A) Breathing apparatus used in mines.
(B) $KO_2 + CO_2 \rightarrow K_2CO_3 + O_2$. '$O_2$' produced is utilised by person.
(C) Both are correct.
(D) None of the above

CSB092

Anomalous properties of Lithium

9. The aqueous solutions of lithium salts are poor conductor of electricity rather than other alkali metals because of
(A) High ionisation energy
(B) High electronegativity
(C) lower ability of $Li^{\oplus}$ ions to polarize water molecules
(D) Higher degree of hydration of $Li^{\oplus}$ ions

CSB080

10. In $LiAlH_4$, metal Al is present in
(A) Anionic part (B) Cationic part
(C) In both anionic and cationic part (D) Neither in cationic nor in anionic part

CSB081

Some Important Compounds of Sodium ($Na_2CO_3.10H_2O$; NaOH; $NaHCO_3$;NaCl) and their uses

11. Which option is correct for Castner Kellner cell.
(A) It is used for preparation of NaOH
(B) In terminal compartment Mercury acts as cathode and carbon rod acts as anode with saturated brine solution.
(C) In central compartment, Fe acts as cathode and amalgam as anode with dilute solution of NaOH as electrolyte
(D) All of these.

CSB090

12. On prolonged exposure to air, sodium finally changes to :-
(A) Na_2CO_3 (B) Na_2O (C) NaOH (D) $NaHCO_3$

CSB088

Paragraph : 13 to 15

$NH_3 + H_2O + CO_2 \longrightarrow X$

$X + NaCl \longrightarrow Y + NH_4Cl$

$Y \xrightarrow{\Delta} Z + H_2O + CO_2$

$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$

$CaO + H_2O \xrightarrow{\Delta} Ca(OH)_2$

$Ca(OH)_2 + NH_4Cl \xrightarrow{\Delta} A + NH_3 + H_2O$

13. Identify Compound 'X'
(A) $(NH_4)HCO_3$ (B) $(NH_4)(HCO_3)_2$ (C) $(NH_4)_2CO_3$ (D) $(NH_4)_2 + HCO_3$

CSB094

14. Identify Compound 'Y'
(A) Na_2CO_3 (B) $NaHCO_3$ (C) Na_2O (D) None of these

CSB095

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15. Identify compound 'Z'.

(A) Na_2O (B) NaO_2 (C) Na_2O_2 (D) Na_2CO_3

CSB096

16. $NaOH(Solid) + CO \xrightarrow{200°C}$ X; product X is

(A) $NaHCO_3$ (B) Na_2CO_3 (C) HCOONa (D) H_2CO_3

CSB089

17. For Solvay's process which option is correct.

(A) The main product is Na_2CO_3

(B) By product of the process is $CaCl_2$

(C) Raw material of the process are NH_3, $CaCO_3$, and saturated brine solution

(D) All of these

CSB091

18. Which option is incorrect for NaOH.

(A) It reacts with CO_2 to give Na_2CO_3 (B) It reacts with ZnO to give Na_2ZnO_2

(C) It reacts with Na_2O to give NaO_2 (D) It reacts with Al_2O_3 to give $NaAlO_2$

CSB093

19. Match the column.

List – I	**List – II**
(I) $Na_2CO_3.10H_2O$	(p) Used in fire extinguisher
(II) NaCl	(q) Used for preparation of ammonia.
(III) NaOH	(r) Manufacturing of Borax from ore Borax.
(IV) $NaHCO_3$	(s) Table salt for domestic purpose

(A) I $\rightarrow$ r, II $\rightarrow$ s, III $\rightarrow$ p, IV $\rightarrow$ q (B) I $\rightarrow$ p, II $\rightarrow$ q, III $\rightarrow$ r, IV $\rightarrow$ s

(C) I $\rightarrow$ q, II $\rightarrow$ p, III $\rightarrow$ s, IV $\rightarrow$ r (D) I $\rightarrow$ r, II $\rightarrow$ s, III $\rightarrow$ q, IV $\rightarrow$ p

CSB098

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Biological importance of Na and K

20. Lye is the commercial name of

(A) RbOH (B) LiOH (C) NaOH (D) CsOH

CSB097

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	C	D	D	C	A	D	C	D	A
Question	11	12	13	14	15	16	17	18	19	20
Answer	D	A	A	B	D	C	D	C	D	C

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RACE Regular Analysis Through Continuous Exercise

The s-Block Elements : RACE-02

Group 2 Elements; Alkaline Earth Metals, Electronic Configuration, Atomic and Ionic Radii, Ionisation Enthalpy, Hydration Enthalpy, Physical Properties

1. The complex formation tendency of alkaline earth metals decreases down the group because
(A) Atomic size increases
(B) Availability of empty d and f-orbitals increases
(C) Nuclear charge to volume ratio increases
(D) All the above

CSB105

2. The correct order of second ionisation potentials (IP) of Ca, Ba and K is
(A) K > Ca > Ba (B) Ba > Ca > K (C) K > Ba > Ca (D) K = Ba = Ca

CSB103

3. If X and Y are the second ionisation potential of alkali metals and alkaline earth metals of same period, then -
(A) X > Y (B) X < Y (C) X = Y (D) X << Y

CSB107

4. How many of the following metal chlorides impart characteristic colour to the oxidising flame.
LiCl, NaCl, KCl, $BeCl_2$, $MgCl_2$, $CaCl_2$, $SrCl_2$, $BaCl_2$

CSB021

5. In how many process coloured product is formed?
(i) Heating of $CaCO_3$
(ii) Na in liquid $NH_3 \rightarrow$ Blue
(iii)Heating of $LiNO_3 \rightarrow NO_2$ (gas)
(iv) $NH_4Cl + Ca(OH)_2$
(v) $Ca(OH)_2 + SO_2$
(vi) $Ca(OH)_2 + Cl_2$

CSB023

6. How many molecules of water are present in formula of epsomite?

CSB026

7. How many s-electrons are present in $Ba^{2\oplus}$?
(Mark your answer as sum of digits)

CSB069

Chemical properties of Alkaline earth metals (Reactivity towards Air, Water, Halogens, Acids, Reducing Nature, Solution in Liquid NH_3)

8. A pair of substances which gives all the same products on reaction with water is
(A) Mg and MgO (B) Sr and SrO
(C) Ca and CaH_2 (D) Be and BeO

CSB108

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9. How many of these compounds can form both acidic and basic products upon heating?
$LiNO_3$, $CaCO_3$, NH_4NO_3, $FeSO_4$, $Mg(OH)_2$

CSB022

10. How many of these compounds are soluble in H_2O?
$CaCO_3$, Na_2CO_3, $BaCO_3$, $BaSO_3$, $Ca(HCO_3)_2$, $AgNO_3$, $(NH_4)_2CO_3$

CSB024

General Characteristic of Compounds of Alkaline Earth Metals (Oxides and Hydroxides, Halides, Salts of Oxo-acids(CO_3^{2-}, SO_4^{2-},NO_3^-))

11. The correct order of basic-strength of oxides of alkaline earth metals is

(A) BeO > MgO > CaO > SrO
(B) SrO > CaO > MgO > BeO
(C) BeO > CaO > MgO > SrO
(D) SrO > MgO > CaO > BeO

CSB106

Some Important Compounds of Calcium (CaO, $Ca(OH)_2$, $CaCO_3$, $CaSO_4.1/2H_2O$)

12.

Column-I	**Column-II**
(A) Hydrolith	(P) Contain Ca
(B) Nitrolim	(Q) Used as a fertilizer
(C) Dolomite	(R) Used to prepare H_2
(D) Pearl's ash	(S) Contain potassium

(A) (A) → S,R ; (B) → P,Q ; (C) → P ; (D) → P
(B) (A) → P, Q; (B) → P,R ; (C) → S ; (D) → P
(C) (A) → P,R ; (B) → P,Q ; (C) → P ; (D) → S
(D) (A) → Q,R ; (B) → P,R ; (C) → P ; (D) → S

Rankers' Academy JEE

CSB061

Biological importance of Magnesium and Calcium

13. The 'Milk of Magnesia' used as an antacid is chemically

(A) $Mg(OH)_2$ (B) MgO (C) $MgCl_2$ (D) MgO + $MgCl_2$

CSB102

14. EDTA is used in the estimation of

(A) $Mg^{2\oplus}$ ions
(B) $Ca^{2\oplus}$ ions
(C) Both $Ca^{2\oplus}$ and $Mg^{2\oplus}$ ions
(D) $Mg^{2\oplus}$ ions but not $Ca^{2\oplus}$ ions

CSB104

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	A	A	6	2	7	10.00	C	3	4
Question	11	12	13	14						
Answer	B	C	A	C						

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